

Enabling High Performance Computing with Pthreads

Multi-core computers can deliver better performance with Pthreads, which is an API for writing multi-threaded applications to boost the performance of a computer. With the examples given in this article, of computing the multiplication of two NxN matrices with a serial application and a parallel application using `p_threads`, you are on your way to developing your own Pthreads apps.



Increasing the clock frequency of a microprocessor to improve the performance was once considered the best option available to industry. But nowadays, due to heat dissipation and energy consumption issues, processor developers have switched to a new model where microprocessors contain multiple processing units called cores. Most modern computing devices have several cores to meet computing requirements. To get high performance out of these cores, developers need to write applications using parallel computing techniques.

This article demonstrates the use of the Pthreads API to achieve high performance by writing multi-threaded applications.

An introduction to the Pthreads API

The Pthreads API is a standard (IEEE Posix 1.3c) application program interface that could potentially be implemented on

many different systems. Pthreads is the standard way multi-threaded support is offered. GNU/Linux implements the Pthreads API by keeping all thread functions and data types in the header file `pthread.h`. Pthread functions are not included in the standard C library. Instead, they are in `libpthread` so we need to compile and link our program as shown below:

```
$ cc -o mythread mythread.c -pthread or $ cc -o mythread
mythread.c -lpthread
```

Developing a multi-threaded application with the Pthreads API

A multi-threaded application has multiple threads executing in a shared address space. A thread is a lightweight process that executes within the same process. All the threads share code and data segments, but each thread will have its own program counter, machine